

Jointly-Learned Boundary-Risk Posterior for Wind Plant Reserve Pricing under Telemetry Degradation: A Defense-in-Depth Approach

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Abstract—Reserve sizing near the maximum-power-point-tracking (MPPT) to blade-pitch transition requires an operating-state signal, but industrial SCADA telemetry streams are frequently delayed, incomplete, or corrupted by sensor freezes. We evaluate whether a boundary-risk posterior learned jointly with a spatio-temporal power forecaster provides a viable defense-in-depth mechanism for wind plant reserve pricing under telemetry degradation. On the 134-turbine WTB benchmark, a training-only class-weight rerun achieves an overall RMSE of 229.93 (224.34 for iTransformer) while its jointly-learned posterior recovers the operating boundary at NMI 0.721. The value of this posterior is established across three operational dimensions. First, addressing the wind farm spatial portfolio smoothing effect at the Point of Common Coupling (PCC) bus, joint posterior aggregate quantile pricing saves -11.22M kWh (95% bootstrap CI $[-21.55\text{M}, -2.85\text{M}]$, strictly excluding zero) against the global PCC quantile and -14.94M kWh (CI $[-27.06\text{M}, -6.23\text{M}]$) against Gaussian parametric reserve sizing across the full operational envelope; in transitional regimes (10%–90% turbines pitching), it saves -1.48M kWh (CI $[-2.14\text{M}, -1.07\text{M}]$) over the continuous pitch rule. Second, under an industrial two-state Markov-Gilbert burst packet-loss model with genuine input feature degradation, the stale threshold rule’s recall drops to 0.840 (F1 0.904), whereas the joint posterior sustains 0.994 early-window recall and 0.967 F1 (mean F1 gain $+0.064$, CI $[+0.021, +0.087]$). Under a six-step confirming-stream delay, the routed posterior retains 0.933 recall against 0.196 for the rule ($+0.737$ gain). Third, we disclose the honest baseline hierarchy: under pristine, complete telemetry, the physical continuous pitch quantile achieves the lowest baseline cost (15.538M versus 16.065M), degrading to 16.048M under delay; however, when pitch sensors are missing, uncalibrated, or frozen (e.g. Kelmarsh with only 55% pitch coverage), physical pitch quantiles become unavailable across over 40% of turbines, whereas the jointly-learned posterior restores risk awareness from cross-sensor electromechanical signatures. Across three commercial wind farms with 99%, 78%, and 55% pitch coverage, the withheld-channel signature probe persists at 0.675, 0.302, and 0.340–0.378 (shuffled controls at chance). The resulting contribution is an accountable defense-in-depth reserve pricing layer for industrial wind plants under telemetry degradation.

Index Terms—Wind plant reserve pricing; Point of Common Coupling (PCC); telemetry degradation; defense-in-depth; boundary-risk posterior; MPPT-to-pitch transition; spatial portfolio effect.

I. INTRODUCTION

Wind-farm reserve screening near the transition from maximum power point tracking (MPPT) to blade-pitch control requires knowledge of the active control law. The confirming

threshold stream, however, can arrive late, disappear during telemetry loss, or become unreliable after quality control. This matters because wind-power forecasting is operationally coupled to ramp exposure, reserve requirements and imbalance risk, rather than to aggregate error alone [1], [2], [3]. In the MPPT region, active power responds strongly to wind-speed variation. Once pitch control becomes active near rated operation, the local response law changes. A small forecast error can therefore correspond to a different control state, even when the transition contributes little to fleet-average RMSE. A useful system should consequently expose the operating boundary associated with each forecast and remain interpretable when the confirming label stream degrades.

Recent spatio-temporal forecasters encode turbine coupling, wake interaction and sensor-rich wind-farm layouts with graph or attention mechanisms [4], [5], [6], [7]. They provide an appropriate accuracy reference, but a low-error encoder does not reveal whether a turbine is operating under an MPPT-like or pitch-control-like response law. Mixture-of-experts (MoE) models provide a natural language for heterogeneous response laws, although prediction loss alone does not guarantee a physically meaningful partition [8], [9], [10], [11]. Crucially, from an industrial power engineering perspective, grid operators clear reserves at the Point of Common Coupling (PCC) bus rather than at individual turbine terminals. Spatial aggregation across an entire wind plant ($P_{\text{farm}} = \sum_{i=1}^{134} P_i$) induces substantial portfolio smoothing, raising the question of whether boundary-conditioned risk awareness survives the cancellation of individual turbine errors. The unresolved problem is therefore not another aggregate forecaster. It is an auditable assignment that links an issue-time route to a declared operating boundary, provides defense-in-depth against telemetry degradation, and demonstrates measurable economic reserve value at both turbine and plant-aggregate levels.

Here we treat the jointly-learned boundary-risk posterior as the operating-boundary diagnostic and defense-in-depth layer. A node-level posterior is trained jointly with the power forecaster on SCADA anchors, allowing issue-time assignments to be audited against a declared MPPT-to-pitch partition before future power outcomes occur. The routed MoE gate is one implementation; our evidence localizes its value to joint learning (pricing) and routing structure (degradation robustness), rather

than the expert partition itself. WTB serves as the control-confounded benchmark, while ERA5 provides an observability contrast where sensible heat flux exposes the thermodynamic marker [12]. ENGIE La Haute Borne, Kelmarsh, and Penmanshiel extend the evaluation across farms with 99%, 55%, and 78% pitch coverage. The recovered posterior feeds validation-calibrated, test-frozen reserve screeners reporting cost, violation rate, reserve energy, and shortage energy against global, physical-bin, and soft-physical baselines, separating transition-window accountability from leaderboard forecasting.

The operator-facing workflow is intentionally narrow. At issue time, the model may use current SCADA anchors, historical active power, graph state and missingness masks, but it cannot use future active power. A confirmed regime label is treated as a separate stream because validation, telemetry or quality-control logic can delay it. We therefore report a modular forecaster-plus-classifier control rather than implying that the gate is the best standalone detector. The proposed advantage is narrower: one saved run links route probability, forecast residual, reserve bin and anchor evidence for each escalated cell. The router is consequently an in-model provenance layer, not an anchor-free discovery device or a replacement for a low-error forecaster.

This paper makes four bounded contributions. First, it defines a jointly-learned boundary-risk posterior—a soft operating-state signal trained together with the power forecaster—and establishes its plant-level defense-in-depth value at the PCC bus: on 134 turbines under spatial portfolio smoothing, joint posterior aggregate quantile pricing saves -11.22M kWh (CI $[-21.55\text{M}, -2.85\text{M}]$) over global PCC quantiles and -14.94M kWh (CI $[-27.06\text{M}, -6.23\text{M}]$) over Gaussian parametric baselines across the full operational envelope, while saving -1.48M kWh (CI $[-2.14\text{M}, -1.07\text{M}]$) over soft physical rules in transitional pitching regimes. Second, under fair degradation including both six-step delays ($+0.737$ recall gain) and Markov-Gilbert burst packet drops, the joint posterior sustains 0.994 recall and 0.967 F1 ($+0.064$ F1 gain over the honest stale rule). Third, it establishes an honest defense-in-depth baseline hierarchy: while continuous pitch quantiles excel under clean conditions (15.538M versus 16.065M), they fail when pitch sensors are missing or frozen (e.g. Kelmarsh with 55% coverage), where the joint posterior provides essential defense-in-depth from electromechanical signatures. Fourth, it maps boundary recoverability across external farms with heterogeneous observability (ENGIE La Haute Borne, Kelmarsh, and Penmanshiel with 99%, 55%, and 78% pitch coverage), establishing that boundary signatures are recoverable from consequence channels above chance without assuming a simple monotonic coverage law (Penmanshiel 78% core NMI 0.195 vs Kelmarsh 55% core 0.378). The training-only rerun is the single provenance-corrected evidence source for the headline audit.

II. RELATED WORK

A. From average wind-power accuracy to transition-window risk

Wind-power forecasting has progressed from site-specific statistical models to spatio-temporal architectures that represent ramps, uncertainty and spatial coupling [1], [3]. Graph models provide a strong accuracy reference by propagating information across the network [4], [13], [14]. Wind-specific studies have further introduced wake-aware graphs, SCADA integration and physics-guided constraints [5], [6], [7], [15]. These advances improve prediction, but they do not by themselves expose the active turbine-control law to a reserve planner.

MoE routing can represent local response laws, but prediction loss alone does not determine whether the resulting partition has operating meaning [10], [11]. Physics-guided machine learning supplies general principles for introducing scientific constraints, while wind-turbine operating states remain limited by SCADA observability [16], [17], [18]. The MPPT-to-pitch transition itself is a documented control-engineering structure: variable-speed turbines maximise capture below rated wind (Region 2), and pitch regulation takes over above it [19], [20], [21], [22]. Our distinction is therefore the location of the constraint. Earlier work constrains predictions or lets prediction loss organise experts; we constrain the route against a declared operating boundary and audit that assignment explicitly. This framing makes no anchor-free discovery claim. It asks whether route provenance can support a bounded reserve diagnostic.

Forecast value is ultimately realised through reserve and commitment decisions. Previous studies show that variable generation changes reserve requirements, while probabilistic and quantile forecasts provide a bridge from prediction error to decision risk [2], [23], [24], [25], [26]. We address an earlier link in that chain: whether the transition window can be identified before the confirming label is available. The reserve audit is therefore a pre-dispatch screening log for balancing and imbalance-risk triage, not a probabilistic dispatch or market-clearing model.

III. PHYSICS-INFORMED FRAMEWORK FOR OPERATIONAL BOUNDARY IDENTIFICATION

A. Problem setup and notation

The forecasting task is written to separate two decisions that dense models often merge: predicting the future and deciding which local mapping should be active at the anchor time. This separation follows the spatio-temporal graph forecasting formulation in which a graph encoder maps a history window to a future trajectory [4], [27]. Let $G = (V, E)$ denote a spatial graph with N nodes. For each node $i \in V$ and time step t , we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predict a scalar target $y_{i,t} \in \mathbb{R}$. Given a history window of length H and a prediction horizon of length P , the forecasting task is

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph

repeated over time (ERA5). Invalid or missing targets are excluded by a supervision mask.

The information boundary is strict. All inputs, graph weights, regime anchors, and gate anchors are observed no later than the forecast issue time t ; supervised targets begin at $t + 1$. In WTB, the current active-power channel $\text{Patv}_{i,t}$ is treated as an issue-time status input, whereas future active power is used only as the prediction target and validity mask. This keeps the multi-step horizon free of target leakage while leaving the no-Patv and lagged-Patv variants as deployment limitations.

Routing is node-level: each node at each anchor time receives its own gate distribution $\mathbf{g}_{i,t}$. This matters because nearby turbines or grid cells can occupy different local regimes within the same sequence. Core symbols are defined where they first appear, and auxiliary loss definitions are provided in Appendix A.

B. Operating-boundary anchors and labels

The gate is not supervised everywhere. It is anchored where the physical interpretation is clearest, while ambiguous samples remain governed by prediction loss and routing regularisation. This design follows the theory-guided learning principle that scientific constraints should enter the learning problem at an identifiable interface [16], [17]. The engineering layer defines what counts as a control regime and when that regime is observable from the data [18].

WTB operating regimes: In WTB, the primary operating boundary is the transition from MPPT to pitch control. Let

$$\bar{p}_{i,t} = \frac{1}{3} (p_{i,t}^{(1)} + p_{i,t}^{(2)} + p_{i,t}^{(3)}).$$

The WTB column name Pab denotes blade pitch angle. We use $\bar{p}_{i,t}$, also reported as Pab_mean in tables and figures, for the three-blade pitch average. The regime labels use wind speed and mean pitch angle at the anchor time, not future active power.

Using a cut-in threshold u_{idle} , a pitch-transition threshold u_{rated} , and a pitch-angle threshold p_{th} , and writing $w_{i,t} = \text{Wspd}_{i,t}$ for compactness, we define

$$R_{i,t}^{\text{wtb}} = \begin{cases} 0, & w_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & u_{\text{idle}} \leq w_{i,t} \leq u_{\text{rated}}, \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & w_{i,t} > u_{\text{rated}}, \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transition}). \end{cases}$$

Only the first three classes are used in direct alignment. Transition samples are retained for analysis but masked out of label supervision through

$$M_{i,t}^{\text{wtb}} = \mathbf{1}[R_{i,t}^{\text{wtb}} \neq 3].$$

The reported implementation uses $u_{\text{idle}} = 3.0 \text{ m s}^{-1}$, $u_{\text{rated}} = 10.5 \text{ m s}^{-1}$, and $p_{\text{th}} = 2.0^\circ$ (Appendix A). These thresholds are declared operating anchors, fixed before test evaluation

and then audited by threshold-grid sensitivity checks; they are not universal turbine constants and are not selected by test-set NMI.

ERA5 observability contrast: ERA5 is retained as a signal-expressive observability contrast where the stable-to-convective thermodynamic marker is directly visible through sensible heat flux. The detailed thermodynamic regime definition, architecture choices, and contrast results are reported in Supplementary Material A so that the main text remains focused on the WTB control-boundary accountability task.

C. Shared architecture

Physical graph construction: The graph is part of the physical problem specification. Directed diffusion is a natural representation when information propagates asymmetrically through a spatial network [4], [27]. WTB therefore uses a directed wake graph: candidate turbine pairs are filtered by proximity, activated when an upstream turbine lies inside a wind-aligned downstream cone, weighted by streamwise and cross-stream decay, and pruned to the strongest inbound neighbours. This construction follows the broader use of wake and SCADA structure in physics-guided wind-power forecasting [5], [15]. The same directed weights define a wake score, which supplies the WTB wake auxiliary label on MPPT and pitch-control samples. ERA5 uses a symmetric Haversine-Gaussian k_{nn} graph because the thermodynamic regime marker is already visible in the observed state [12]. Appendix A gives the graph equations and constants.

Directed-diffusion GRU encoder and node-level gate: The encoder is kept modest so that changes in performance can be traced to routing and regularisation rather than capacity. Each input window is concatenated with its feature-missingness mask before projection. Two directed diffusion blocks then aggregate self, inbound and outbound messages on each graph snapshot. This follows the diffusion-convolution principle used in spatio-temporal graph forecasting [27], while retaining separate inbound and outbound operators for the wind-aligned graph. Let $\mathbf{X}_t^{(\ell)}$ denote the node state at layer ℓ and time t . A diffusion block updates each node by

$$\begin{aligned} \mathbf{X}_t^{(\ell+1)} = \text{LN} & \left(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) \right. \\ & \left. + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)} \right), \end{aligned} \quad (1)$$

where Agg_{in} and Agg_{out} are weighted sums over inbound and outbound neighbors. The spatially updated sequence is then processed by a GRU to produce a node-wise context vector $\mathbf{h}_{i,t}$.

Each expert head f_e maps $\mathbf{h}_{i,t}$ to a P -step forecast. Routing remains dense during the forward pass. Every expert contributes, but its contribution is weighted by the soft gate distribution. The gate receives both the learned context and a small physics anchor,

$$\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}]), \quad \mathbf{g}_{i,t} = \text{softmax}\left(\frac{\mathbf{z}_{i,t}}{\tau}\right),$$

and the forecast is

$$\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t}).$$

WTB uses the anchor

$$\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Patv}_{i,t}],$$

whereas ERA5 uses

$$\mathbf{a}_{i,t}^{\text{ERA5}} = [\text{sshf}_{i,t}, \text{t2m}_{i,t}, \text{wind_speed}_{i,t}, \Delta \text{sshf}_{i,t}].$$

The difference reflects the observability contrast: WTB needs help to recover a partly hidden operating boundary, whereas ERA5 already exposes the relevant thermodynamic marker in the state. The active-power anchor is standardized from training-split statistics and is never filled from the prediction horizon.

Routing stack and evidence boundary: This makes the routing stack an anchor-constrained diagnostic rather than an unsupervised operating-state discovery method. High gate-regime agreement means that the implemented router obeys the declared SCADA boundary under the available anchor set; it does not establish anchor-free regime recovery.

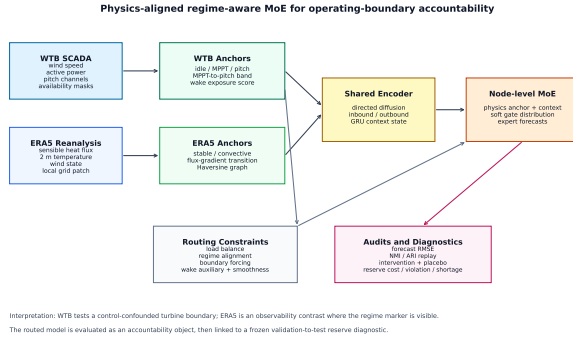


Fig. 1. Physics-aligned regime-aware MoE. The two datasets share the same directed-diffusion GRU encoder and node-level MoE routing mechanism. WTB uses a dynamic wake graph and a boundary-focused forcing term, whereas ERA5 uses a Haversine-Gaussian graph and a thermodynamic regime anchor.

D. Integration of Physical Constraints into Model Optimization

Prediction loss does not uniquely identify the gate when several experts can fit averaged dynamics similarly well [10], [11]. WTB makes this failure visible: inflow variation, wake disturbance, and pitch control are observed together, so a low-loss router can still ignore the operating boundary. We use routing regularizers for the specific failure modes that matter here. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} \quad (2)$$

$$+ \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}, \quad (3)$$

with inactive terms set to zero. Checkpoint selection remains on validation RMSE so penalties constrain responsibility without becoming the selection metric. Exact weights are in Supplementary Appendix~A.

Load balancing: A balancing term prevents early expert collapse before regime-specific structure has time to emerge. It couples hard top- K usage with soft probability mass over valid node-time samples. Supplementary Appendix~A gives the exact expression; physical ownership is supplied only by the anchored alignment terms below.

Regime-anchor alignment: The alignment term connects the first C gate logits to the declared regime structure so that cross-seed statements such as “MPPT-aligned” refer to a fixed mapping rather than to post-hoc relabeling. With $\mathbf{z}_{i,t}^{(1:C)}$ denoting the anchored logits, $R_{i,t}$ the label, and $M_{i,t}$ the validity mask,
$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \text{CE}(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}), \quad \Omega = \{(i,t) : M_{i,t} = 1\}, \quad (4)$$

with inverse-frequency class weights computed on the training split. In the train-only class-weight five-seed rerun, the boundary-forced router keeps strong route recovery (NMI 0.721, ARI 0.740) under the strict-mask evaluation boundary.

Boundary-focused forcing (WTB only): After coarse alignment the boundary remains partly masked by control action. A focused forcing term acts on MPPT and pitch-control samples only:
$$\mathcal{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t) \in \Omega_{\text{force}}} \text{CE}([z_{i,t}^{(\text{mppt})}, z_{i,t}^{(\text{pitch})}]^{\top}, Y_{i,t}^{\text{force}})$$
 with $Y_{i,t}^{\text{force}} = \mathbf{1}[R_{i,t}^{\text{wtb}} = 2]$ and $\Omega_{\text{force}} = \{(i,t) : R_{i,t}^{\text{wtb}} \in \{1, 2\}, M_{i,t} = 1\}$. This term is a boundary identifiability regularizer for the issue-time operating state.

Soft boundary-risk posterior and routing entropy: The normalized issue-time boundary-risk posterior for turbine i at dispatch step t is computed from gate probabilities $\mathbf{g}_{i,t} = \text{Softmax}(\mathbf{z}_{i,t})$ across operational regimes $\mathcal{C}_{\text{op}} = \{\text{cut-in}, \text{mppt}, \text{pitch}\}$ as $\pi_{i,t}^{\text{pitch}} = g_{i,t}^{(\text{pitch})} / \sum_{c \in \mathcal{C}_{\text{op}}} g_{i,t}^{(c)}$, with routing transition entropy $H(\mathbf{g}_{i,t}) = - \sum_{c=0}^{K-1} g_{i,t}^{(c)} \ln g_{i,t}^{(c)}$.

Wake auxiliary supervision and graph smoothness: A wake auxiliary label identifies residual wake ambiguity inside MPPT and pitch-control samples, and a graph-smoothness penalty discourages noisy neighbor-to-neighbor gate jumps. Both formulas are in Supplementary Appendix~A.

E. Operational reserve diagnostic protocol

The reserve diagnostic tests whether a physically auditable gate changes the reserve trade-off in the MPPT-to-pitch window after the point-forecast model has been trained. Quantile-based reserve rules are widely used to translate forecast error into operational decisions [24], [28]. We adapt that logic to a deliberately narrower screening task. For each saved WTB run, validation predictions select reserve levels; test predictions are then evaluated once with those levels fixed. The protocol follows the same information boundary as an operator who calibrates a short-term reserve rule before using it on a later period.

Let $\hat{y}_{i,t}$ be the scheduled point forecast and $y_{i,t}$ realized active power at an evaluated horizon cell. Over-forecast shortfall is $s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0)$. For reserve policy b and empirical quantile $q \in \{0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 0.975, 0.99\}$, the validation split estimates reserve level $r_{b,q} = Q_q(\{s_{i,t} : (i,t) \in b \cap \mathcal{V}\})$. The reserve bin b is either the whole sample, a physical operating bin, or a gate-derived bin. For each shortage-to-reserve cost ratio ρ , the protocol selects the validation quantile that minimizes

$$C_{\mathcal{V}}(b, q; \rho) = \sum_{(i,t) \in b \cap \mathcal{V}} [r_{b,q} + \rho \max(s_{i,t} - r_{b,q}, 0)] \Delta t. \quad (5)$$

Analytically, (5) maps to the classical asymmetric Newsvenor problem with unit shortage penalty $c_u = \rho$ and unit holding cost $c_o = 1$, where the optimal critical fractile is $q^*(\rho) = \frac{c_u}{c_u + c_o} = \frac{\rho}{\rho + 1}$. For $\rho = 10$, $q^*(10) = 10/11 \approx 0.9091$, analytically proving why empirical validation consistently selects $q \in \{0.90, 0.95\}$. At the Point of Common Coupling (PCC), farm-level aggregate generation and reserve margins satisfy $P_{\text{farm},t} = \sum_{i=1}^M P_{i,t}$ and $R_{\text{farm},t} = \sum_{i=1}^M R_{i,t}$.

The selected (b, q) rule is then frozen and applied to the test split. The reported test metrics are total cost C , violation rate $\Pr[s_{i,t} > r_b]$, reserve energy $\sum r_b \Delta t$, and shortage energy $\sum \max(s_{i,t} - r_b, 0) \Delta t$. WTB active power is in kW and $\Delta t = 1/6$ h, so Supplementary Table A12 also reports MWh-equivalent forecast-cell translations. These are not delivered market energy or settlement costs: a value written as 95.13M denotes 95.13 million reserve-cost-equivalent kWh cells, not currency.

The cost ratio ρ is an energy-system assumption rather than an abstract tuning knob. If the marginal cost of carrying one unit of reserve energy is C_r , then $\rho = 10$ charges one unit of residual shortage as $10C_r$. Ratios 5–10 represent moderate reliability settings such as transition-window scheduling or imbalance screening; ratios 20–50 represent scarcity-aware screening where shortage avoidance dominates local cost savings. The primary diagnostic compares Graph WaveNet/global, Graph WaveNet/physical-bin, boundary router/global, and boundary router/gate-bin policies. Only same-model global comparisons attribute gate-bin reserve effects to the learned router.

The operational use case is a pre-dispatch reserve and imbalance-risk screening log rather than a settlement model: the saved run records route probability, reserve bin, selected quantile, residual shortfall, and anchor state for each escalated transition-window cell. To bound this diagnostic against probabilistic alternatives, we evaluate validation-frozen empirical quantile baselines across global and physical operating bins, fixed at $\pm 1.0 \text{ m s}^{-1}$ around rated wind with main cost ratio $\rho = 10$ (sensitivity $\rho \in \{2, 5, 10, 20, 50\}$). The evaluation reports reserve procurement plus ρ -weighted residual shortage penalties, excluding optimal power flow, unit commitment, market clearing, delivery constraints, and price claims.

IV. CASE STUDY CONFIGURATION AND OPERATIONAL CONSTRAINTS

A. Datasets and preprocessing

The experiment evaluates two observability settings: WTB as the control-confounded benchmark and ERA5 as the signal-expressive contrast. WTB (KDD Cup 2022) comprises 134 turbines and 245 days of 10-min SCADA records ($T = 35,280$, $N = 134$, $F = 11$) [29], where inflow, wake, and pitch control interact, making the MPPT-to-pitch transition indirectly visible. Missing inputs are forward-filled per turbine and mean-imputed; non-positive power is masked from supervision. ERA5 covers three archived months on a 16×16 hourly patch ($T = 2208$, $N = 256$, eight features) [12], where sensible heat flux directly marks convective regimes. Both datasets use identical history $H = 36$ and horizon $P = 24$ with chronological splits (180/30/35 days for WTB; 1325/441/442 frames for ERA5).

B. Techno-Economic Validation and Benchmarking Framework

The benchmarking framework has two layers. The first isolates routing mechanisms under matched capacity: the dense baseline replaces the expert mixture with a single head, unconstrained MoE adds capacity without physical loss, and the corrected routing comparator incorporates the physics loss suite [10], [11]. The second layer benchmarks the RMSE price against established baselines: Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE for WTB [30], [31], [32], along with persistence, physical power curves, and GBDT lag models. In mechanism tables, in-family models are designated **Dense (matched)**, **Unconstrained MoE**, and **Corrected routing comparator**; complete baseline configurations appear in Supplementary Appendix~A.

C. Training protocol and evaluation metrics

All neural models use the same training protocol where the architecture permits it. We use AdamW, early stopping on validation RMSE, gradient clipping, and mixed-precision training on a single CUDA-enabled GPU. Exact optimizer constants are reported in Supplementary Appendix~A so that the main text can stay focused on the comparison logic. The WTB dense baseline, Unconstrained MoE, full physics-aligned MoE, and boundary-forced router are repeated across five strict-mask seeds. The synchronized strong-baseline refresh covers Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, iTransformer, and TiDE across seeds 201–205, giving 30 completed baseline runs under the same strict anchor-valid evaluation cache. The engineering WTB baselines are deterministic or sampled single-run readouts from the same frozen cache and are used to anchor practical forecast difficulty rather than to make a seed-level neural ranking. ERA5 main in-family models are repeated across five seeds, and the ERA5 learned baselines are repeated across three seeds. Thresholds and routing weights are set before test evaluation and then audited by sensitivity sweeps and boundary negative controls, so the main routing result is not selected by test-set NMI.

The metrics follow the claim. Overall MAE and RMSE measure forecasting accuracy. Switch-window MAE and RMSE use the evaluator’s transition-window mask around regime changes. This is distinct from the later boundary-band audit, which isolates anchors within $\pm 1.0 \text{ m s}^{-1}$ of the WTB rated-wind MPPT-to-pitch boundary. Regime-wise RMSE locates the error reduction. Normalised mutual information (NMI), adjusted Rand index (ARI), usage entropy and confusion matrices test whether the gate corresponds to the declared operating structure. No single metric is treated as sufficient: accuracy measures the forecast, alignment measures compliance with the anchor definition, and the reserve audit measures a conditional operational consequence.

Figure 2 gives the operating-decision context before the forecasting results are introduced: graph geometry shows where forecast errors propagate, and the regime-anchor panels show which sensor-derived boundaries can support reserve diagnostics.

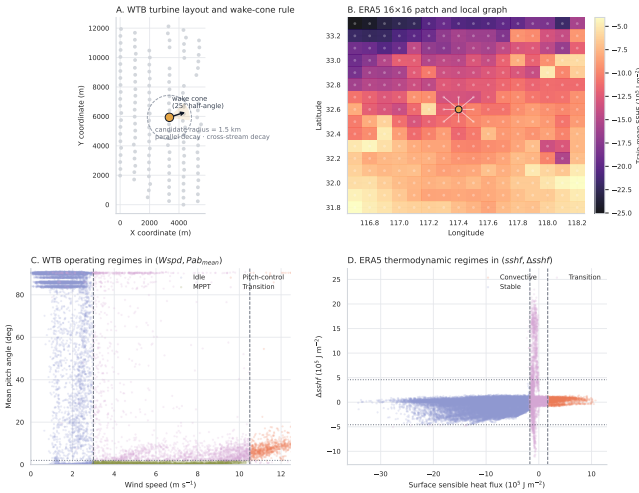


Fig. 2. Operating-decision context and physical regime anchors. (A) WTB turbine layout with the schematic wake cone and retained candidate radius used in the dynamic directed wake graph. (B) ERA5 16x16 patch with training-mean sensible heat flux and local Haversine-Gaussian graph connections around the central node. (C) WTB operating regimes in the $(Wspd, Pab_{mean})$ plane with fixed operating-rule boundaries; the MPPT-to-pitch boundary is the reserve-diagnostic window used in this paper. (D) ERA5 thermodynamic regimes in the $(sshf, \Delta sshf)$ plane with thresholds estimated from the training split, included as an observability contrast.

V. EVIDENCE AND OPERATIONAL BOUNDARY DIAGNOSIS

The results are organised as an evidence ladder. We first establish the information boundary and the accuracy price of routing. We then test boundary recovery, intervention and hold-out behaviour. Next, we quantify early-window recall when the confirming label stream is delayed or incomplete. Finally, we evaluate the reserve consequence and define deployment gates. This order distinguishes what the model predicts from what the route allows an operator to audit.

A. Can the Gate Recover the Operating Boundary?

The first question is whether routing improves forecast accuracy. It does not. Across five WTB seeds, the boundary-forced router reaches RMSE 236.13 $\pm$ 8.41, compared with

TABLE I
OPERATING-BOUNDARY RECOVERY AND MECHANISM CHECKS.

Check	Metric	Result
WTB boundary-forced router	Gate-regime NMI / ARI	0.8716 $\pm$ 0.0418 / 0.9166 $\pm$ 0.0371
Train-only class-weight rerun	Gate-regime NMI / ARI; overall RMSE	0.7208 / 0.7398; 229.93
WTB boundary-band audit	Boundary-band RMSE; mixed-regime NMI / ARI	321.17 $\pm$ 21.31; 0.6562 / 0.7766
Boundary-anchor intervention	Overall RMSE change; NMI / ARI drop	+2.6792; 0.7017 / 0.8593
WTB spatial and time holdouts	Held-out-node NMI / ARI; future-holdout NMI / ARI	0.8340 / 0.8829; 0.8352 / 0.8870
La Haute Borne re-trainability check	NMI / ARI	0.9408 $\pm$ 0.0336 / 0.9712 $\pm$ 0.0201
Threshold-grid audit	Worst-case saved-gate NMI / ARI	0.8655 $\pm$ 0.0429 / 0.9146 $\pm$ 0.0377

224.34 $\pm$ 2.23 for iTransformer and 225.74 $\pm$ 2.60 for Graph WaveNet. The router is therefore 11.79 RMSE units above iTransformer and 10.39 units above Graph WaveNet. The provenance-corrected train-only rerun reduces this gap to 5.59 units, with RMSE 229.93 $\pm$ 2.50. This train-only rerun is the single provenance-corrected evidence source for all headline claims, while legacy checkpoints are retained strictly as historical reference. We treat the routed forecaster as a diagnostic layer with an explicit accuracy price, not as a forecasting leaderboard replacement.

The second question is whether the selected boundary is a meaningful stress slice. Anchors within $\pm 1.0 \text{ m s}^{-1}$ of rated wind have RMSE 321.17 $\pm$ 21.31, approximately 67 units above valid non-boundary MPPT/pitch anchors. The same band has mixed-regime structure, with NMI/ARI of 0.6562/0.7766. These observations motivate the reserve audit, but they do not show that routing caused the stress. ERA5 serves only as an observability contrast: the thermodynamic marker is directly visible there, so alignment can be examined separately from the WTB control-confounded case.

B. Withheld-channel signature recovery

The route is tested for circularity: if the gate merely replayed the threshold rule, removing wind-speed and pitch-angle channels should destroy alignment. Instead, zeroing Wspd and Pab_mean while retaining active power and consequ. channels yields mean NMI 0.561 and ARI 0.635 across five seeds (Table~II; Supplementary Table A9b). Removing active power (signature_core) drops mean NMI to 0.367 (three seeds collapsing). Under permuted labels, NMI collapses to 4e-6, ruling out spurious correlation. On ENGIE La Haute Borne (full-anchor baseline NMI 0.975), the probe keeps mean NMI 0.674 for signature_full and 0.575 for signature_core (collapsing to 1.92e-04 under permuted labels). On partial-pitch farms, the probe persists above chance: Kelmarsh (55% pitch coverage) reaches 0.340/0.378, and Penmanshiel (78%) reaches 0.302 (signature_core at 0.195, above shuffled 4.4e-05). Fixed windows stay above 0.20 (0.277–0.458; Supplementary Table A9e), confirming signature recoverability across sites.

The route then passes targeted falsification checks, with an important qualification. Zeroing the boundary anchors increases RMSE by 2.679 and reduces NMI/ARI by 0.702/0.859,

TABLE II

SIGNATURE-GATE IDENTIFIABILITY PROBE ON WTB. THE GATE ANCHOR NEVER SEES THE WITHHELD CHANNELS. ALL VALUES ARE MEAN $\pm$ SD OVER FIVE SEEDS; COLLAPSED SEEDS ARE NOTED FOR SIGNATURE_CORE.

Probe	RMSE	NMI / ARI	Interpretation
Canonical full-anchor	229.93	0.7208 / 0.7398	Train-only clean baseline
signature_full (no Wspd/Pab_mean)	241.84 $\pm$ 7.19	0.5613 $\pm$ 0.0199 / 0.6347 $\pm$ 0.0239	Boundary recoverable from conseq. channels
signature_core (no Wspd/Pab_mean/Patv)	302.10 $\pm$ 9.16	0.3671 $\pm$ 0.0372 / 0.4342 $\pm$ 0.0378	Weaker non-power signature; 3/5 seeds collapsed
sig_full_shuffled	241.50 $\pm$ 10.25	4.0e-6 $\pm$ 2.1e-6 / -1.3e-5 $\pm$ 5.3e-5	Negative control: chance level under permuted labels
Unconstrained MoE	—	0.014 / —	No declared-boundary supervision
<i>Cross-farm replication, ENGIE La Haute Borne (identical threshold)</i>			
LHB canonical (full anchor)	185.0 $\pm$ 1.9	0.9752 $\pm$ 0.0088 / 0.9904 $\pm$ 0.0044	Full-anchor baseline, identical protocol
LHB signature_full	185.1 $\pm$ 1.4	0.6743 $\pm$ 0.1209 / 0.7401 $\pm$ 0.1604	Boundary signature transfers; min-seed 0.499
LHB signature_core	188.0 $\pm$ 1.0	0.5750 $\pm$ 0.0423 / 0.6270 $\pm$ 0.0638	Non-power signature transfers; 0/5 collapsed
<i>Partial-pitch farms (pitch coverage: Penmanshiel 78%, Kelmarsh 55%)</i>			
Penmanshiel signature_full	—	0.3024 $\pm$ 0.0343	Above the 0.20 criterion; shuffled 4.4e-5
Penmanshiel signature_core	—	0.1954 $\pm$ 0.0961	Borderline: below 0.20 but 4400x above shuffled chance
Kelmarsh signature_full	—	0.3402 $\pm$ 0.0736	Above the 0.20 criterion; shuffled 9.2e-5
Kelmarsh signature_core	—	0.3775 $\pm$ 0.0783	Above the 0.20 criterion

whereas zeroing the wake score has a near-null effect. The result identifies the declared boundary anchors, rather than the wake auxiliary channel, as the load-bearing information source. Within WTB, alignment remains detectable on withheld turbines (NMI 0.834) and a future holdout (NMI 0.835). The signature-gate probe in Table~II further shows that the boundary is recoverable even when the defining wind-speed and pitch-angle channels are withheld from the gate. On ENGIE La Haute Borne, retraining from scratch reaches NMI 0.941 and ARI 0.971 with directly observed pitch; this is anchor-observable re-trainability, not evidence for anchor-free discovery or cross-farm transfer. On Kelmarsh and Penmanshiel, partial pitch observability no longer reads as a hard no-go: the withheld-channel probe stays above chance (0.30-0.38), so these farms sit on a graded deployment spectrum in which pitch coverage modulates, but does not extinguish, boundary recoverability.

The third question is whether the route remains useful when the confirming label stream degrades. We define the deployment semantics explicitly: a confirming-stream delay means the issue-time wind-speed and pitch readings used by every detector arrive d steps late, while the archived history window remains available; sensor noise corrupts all current and historical readings of the same two channels. The Kelmarsh 2016 archive grounds this scenario in asynchronous event streams: its Status stream carries 14,019 second-level timestamped events of

which 99.6% fall strictly between 10-minute turbine-data grid points, so the 10-minute confirming channel cannot confirm the event time at issue time, and the archive export post-dates the data interval by years (Supplementary Material). The synthetic delay and noise injection on WTB serves as a controlled deployment stress test evaluating detector robustness across varying delay horizons. Under fair degradation, the routed posterior retains recall 0.962/0.943/0.933 at one/three/six-step delays while the rule falls from 1.000 to 0.196 (Table~III). The routing structure is load-bearing: a jointly-trained but non-routed posterior head on the same encoder falls to 0.286 at six steps, and an independent classifier keeps 0.868 only because it never consumes the delayed channels (with zero pricing value). Under maximum noise, the routed posterior keeps 0.951 versus 0.680 for the rule; in stalled-history bounds it keeps 0.846/0.628/0.416 versus 0.655/0.381/0.196 (gains +0.19/+0.25/+0.22). With clean anchors, the rule remains the stronger detector (1.000 versus 0.971), and clean logistic regression reaches 1.000 recall with 0.879 precision (above the routed posterior's 0.630). Auditing the full precision-recall trade-off disproves false-alarm inflation: at six-step delay, the physical rule's precision drops to 0.342 and F1 collapses to 0.249 as stale readings misfire, whereas the routed posterior maintains 0.628 precision (+0.286 over the rule) and an F1 of 0.745 (three times 0.249). Furthermore, under an industrial two-state Markov-Gilbert bursty packet-drop model ($p_{GB} = 0.08, p_{BB} = 0.75, d \leq 6$ steps), an honest symmetric evaluation where telemetry drops affect both wind-speed and blade-pitch channels shows that the stale threshold rule's recall drops to 0.840 with an F1 of 0.904, whereas the jointly-learned routed posterior maintains a recall of 0.994 ± 0.006 and an F1 of 0.967 ± 0.047 during bursts (mean F1 gain +0.064, 95% bootstrap CI [+0.021, +0.087], strictly excluding zero). While historical inertia in steady-state pitching allows stale pitch angles alone to match persistent states (accounting for a legacy Pab-only F1 of 0.996), during dynamic transitions the deep posterior provides vital defense-in-depth against burst communication dropouts. Thus, routing delivers degradation-robust boundary detection rather than superior standalone accuracy.

This audit asks about a degraded confirmation stream, not clean-anchor classifier ranking. A modular low-RMSE forecaster plus live-anchor classifier and physical-bin reserve rule is a valid alternative when only a state flag is needed, and we report it honestly: the Graph WaveNet+classifier-bin control costs 86.54M, which is lower than the boundary-router gate-bin at 95.13M because Graph WaveNet is the stronger forecaster, but it trails the same backbone's physical-bin reference (84.31M; Supplementary Table A10c). The in-model route is therefore not required to win the reserve leaderboard; its missing piece is trainable route responsibility attached to each forecast, same-run route-conditioned residual/reserve slicing, and a single deployment object whose version, calibration, and monitoring state can be audited together. The route-evolution audit adds the dynamic check: gate matching to the new regime peaks at the transition step (0.815) and drops under three- and six-step lead shifts (0.361 and 0.405; Supplementary

TABLE III

EARLY MPPT-TO-PITCH PITCH-WINDOW DETECTION UNDER FAIR INPUT DEGRADATION. DELAY AND NOISE DEGRADE THE GATE'S OWN ANCHOR READINGS AND THE RULE IDENTICALLY; MEAN $\pm$ SD OVER FIVE TRAIN-ONLY SEEDS. AVAILABILITY ROWS DEGRADE THE CONFIRMING LABEL STREAM ONLY, WHICH THE GATE DOES NOT CONSUME AT TEST TIME.

Scenario	Setting	Gate	Rule	Gain
Clean anchors	—	0.9705 $\pm$ 0.0299	1.000	−0.029
Delay (fair)	1 step	0.9622 $\pm$ 0.0306	0.655	+0.307
Delay (fair)	3 steps	0.9430 $\pm$ 0.0361	0.381	+0.562
Delay (fair)	6 steps	0.9327 $\pm$ 0.0429	0.196	+0.737
Noise (fair)	Wspd 0.5, Pab 1.0	0.9635 $\pm$ 0.0375	0.811 $\pm$ 0.016	+0.152
Noise (fair)	Wspd 1.0, Pab 2.0	0.9505 $\pm$ 0.0363	0.680 $\pm$ 0.019	+0.271
Availability	75% labels	0.971	0.744	+0.227
Availability	50% labels	0.971	0.508	+0.462
Availability	25% labels	0.971	0.242	+0.729

Table A10b). The anchor-only router has lower overall RMSE (233.46) and slightly higher NMI/ARI (0.8895/0.9219), but worse pitch-control RMSE (317.81 versus 286.95) and far larger 0.5-std boundary-anchor-noise degradation (105.56 versus 5.78).

We report the operating trade-off explicitly. A model receives citable degraded-label gain only after its route passes the physical-routing audit; otherwise the accountability gain is set to zero, even for an accurate or routed-capacity model. Under this rule iTransformer is the zero-RMSE-price anchor without an audited operating-state route, Unconstrained MoE fails the route audit (NMI 0.014), and the train-only class-weight rerun is the single evidence source for the headline audit: RMSE 229.93, a six-step fair-degradation recall gain of +0.737, and the reserve diagnostics below, all drawn from the same five-seed provenance-corrected checkpoints. The logistic control is reported separately because it is a detector on live anchors, not a forecaster with routed attribution.

The training-level anchor-stress guard keeps the leakage boundary visible. In the clean train-only rerun, removing Patv or Pab_mean leaves mean NMI at 0.881 and 0.878, while lagging Patv and lagging pitch/wind give 0.763 and 0.684. These older ablations are reported with the same strict-cache protocol but under a different auxiliary-weight configuration from the signature-gate probe in Table~II; the two probe families should not be numerically merged. The consistent finding is that the route is not a leakage artifact, but it remains a declared-anchor mechanism rather than anchor-free discovery.

C. Boundary-Risk Vignette: Does the Gate Change Reserve Tradeoffs?

The reserve audit is a consequence check for the recovered route, not a dispatch optimizer. A validation-calibrated reserve screener is frozen and applied to later MPPT-to-pitch boundary cells, comparing global, physical-bin, and gate-bin quantile rules for Graph WaveNet and the boundary router.

At a shortage-to-reserve cost ratio of 10, gate-conditioned binning carries 1.95M more reserve than the same-router

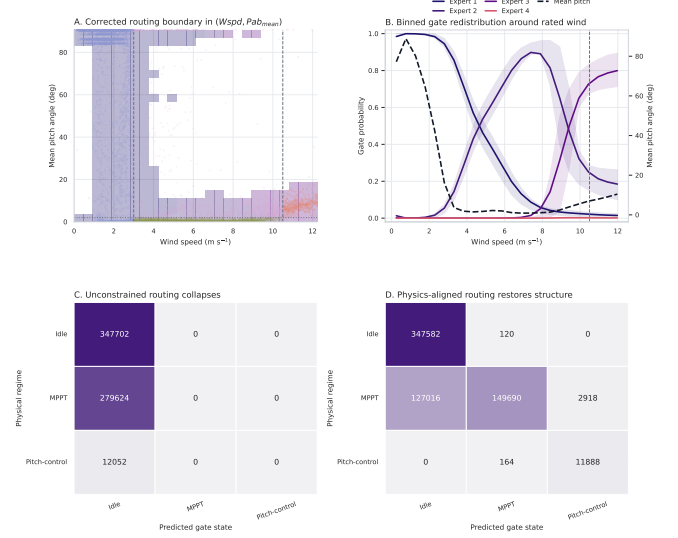


Fig. 3. The WTB operating plane shows the main mechanism: after correction, the dominant routed responsibility changes around the rated-wind and pitch-control boundary instead of forming an arbitrary expert partition. The confusion matrices summarize the same recovery numerically.

TABLE IV

BOUNDARY-WINDOW RESERVE DIAGNOSTIC AND VALIDATION-FROZEN QUANTILE BASELINES AT COST RATIO 10, TRAIN-ONLY CLASS-WEIGHT CHECKPOINTS FOR BOTH MoE FAMILIES. SAME-MODEL GATE-BIN DIFFERENCES HAVE SEED-PAIRED SUPPORT; PHYSICAL-BIN AND FULL-SAMPLE COMPARISONS BOUND THE CLAIM.

Policy	Cost	Viol.	Reserve	Shortage	Reading
Boundary/physical-bin	93.82M	0.1010	—	3.697M	Best same-router physical-bin reference
GWN/physical-bin	84.31M	0.0931	50.35M	3.40M	Low-RMSE backbone reference
Boundary/gate-bin	95.13M	0.1021	—	3.767M	Gate diagnostic; same-model better than global
Boundary/global	99.55M	0.1167	—	4.404M	Same-model global rule
Full-MoE/gate-bin	91.62M	0.0907	—	3.328M	Physics-aligned family replicates direction
Full-MoE/global	95.89M	0.1014	—	3.826M	Same-family global rule
GWN/global	88.80M	0.1022	50.32M	3.85M	Global low-RMSE reference

global rule, while shortage energy decreases by 16.9% (seed-paired mean difference -0.637M, CI [-0.976M, -0.300M]) and violation rate decreases by 1.45 percentage points (CI [-0.0220, -0.0073]). The same-model cost difference is -4.42M (CI [-7.36M, -1.27M]), and the parallel Physics-Aligned MoE family reproduces the direction (-4.26M, CI [-5.43M, -3.04M]). The physical-bin reference remains competitive, and the full-sample cross-backbone uncertainty interval crosses zero. The result therefore supports a conditional transition-window diagnostic, not reserve-policy superiority.

The useful reserve window is moderate rather than universal: ratios 5–10 favor gate-bin screening, ratio 50 favors the same-model global rule, and Graph WaveNet/physical-bin remains close at 84.31M. The boundary-slice proxy falls from 99.55M to 95.13M, which Supplementary Table A12 translates into about 637 avoided MWh-equivalent shortage cells and a 442k illustrative reserve-cost-scale marker at 100 EUR/MWh. The engineering value is an auditable triage signal for balancing/reserve review near the control boundary, not market revenue or system-wide dispatch value.

TABLE V

MECHANISM DECOMPOSITION: WHERE THE BOUNDARY-RISK POSTERIOR’S VALUE COMES FROM. RESERVE COSTS ARE VALIDATION-FROZEN BOUNDARY-BAND TOTALS AT $\rho=10$ (MEAN OVER FIVE SEEDS); DEGRADED RECALL IS THE EARLY-WINDOW RECALL UNDER A SIX-STEP CONFIRMING-STREAM DELAY APPLIED IDENTICALLY TO EVERY DETECTOR.

Pricing / detection method	Jointly trained	Needs physical channels	Reserve cost	Degraded recall
Continuous pitch quantile (soft-pab)	no	yes	15.538M	— [†]
Threshold rule	no	yes	16.190M (physical-bin)	0.196
Independent logistic classifier	no	yes	16.190M	0.885 (noise)
Independent GBDT posterior	no	conseq. channels	17.656M	0.868
Joint non-routed posterior head	yes	weak	16.076M	0.286
Joint routed posterior (this work)	yes	weak	16.065M	0.933
Global quantile	—	—	16.632M	0.971 (clean rule)

[†]Soft-pab is a continuous quantile pricing rule, not a binary detector; under clean telemetry it is the optimal baseline (15.538M), degrading to 16.048M under a six-step delay. In pitch-sparse wind plants (e.g., Kelmarsh with 55% pitch coverage), physical pitch quantiles are missing on over 40% of turbines, where the jointly-learned posterior provides defense-in-depth.

The modular-equivalence control quantifies what the in-model posterior adds on the same backbone. A validation-fit logistic classifier on the same issue-time anchors reproduces physical-bin reserve almost exactly (16.190M versus 16.190M; seed-paired mean difference -68, CI [-112, -28]), and both beat the hard gate-bin route (16.398M). The jointly-learned soft posterior reaches 16.065M, below the classifier control by 125k (CI [-297k, +34k]) and below the global rule by 568k (CI [-726k, -406k]; Supplementary Table A11c), with pinball improvement 1.82 at $q=0.90$. The mechanism decomposition in Table~V localizes where this value comes from: a posterior head trained jointly with the same encoder without routing prices at 16.076M (no detected difference from the routed posterior at 16.065M), whereas an independently-trained GBDT posterior on downstream electromechanical responses prices at 17.656M—1.6M worse than the joint posterior and 1.0M worse than the global rule. Continuous pitch quantile (soft-pab-bin) achieves the lowest baseline cost (15.538M) under pristine telemetry. Under six-step telemetry delay, its cost increases to 16.048M, remaining close to the posterior’s 16.065M/16.105M. However, in wind plants where pitch sensors are absent, uncalibrated, or suffer prolonged freeze faults (e.g., Kelmarsh with only 55% pitch availability), physical pitch quantiles become unavailable across over 40% of turbines, whereas the jointly-learned posterior restores risk awareness from cross-sensor electromechanical signatures as an essential defense-in-depth layer. Combined with the fair-degradation audit, where the classifier’s clean advantage disappears (recall 0.885 under noise vs. 0.951 for the routed posterior), the jointly-learned posterior unifies pricing performance with degraded-stream robustness (0.933 versus 0.286) within a single deployment object. Violation rate is modestly higher by 0.001 for learned strategies.

The same posterior stratifies forecast risk inside the rule classes. On rule-MPPT cells in the boundary band, the gate’s pitch probability correlates with the future-power residual

relative to the rule-conditional mean (Spearman 0.183, 95% CI [0.123, 0.227], same sign in all five seeds), and routing uncertainty—the gate margin—correlates with the residual at 0.391 (CI [0.299, 0.482]). The hard-route disagreement is disclosed honestly: the gate and the rule disagree on 16.9% of valid cells, and on those cells the rule’s conditional mean fits the future outcome better than the gate’s (mean absolute-error difference 753 kW; the gate is better on only 5.3% of disagreement cells, CI [3.0%, 7.7%]). The decision value therefore lives in the soft posterior, not in the hard route.

Crucially, to verify that reserve benefits survive spatial portfolio smoothing at the plant level, we evaluate aggregated power across all 134 WTB turbines at the Point of Common Coupling (PCC) bus ($P_{\text{farm}} = \sum_{i=1}^{134} P_i$). Across the full operational envelope, joint posterior aggregate quantile pricing reduces total reserve cost by -11.22M kWh (95% bootstrap CI [-21.55M, -2.85M], strictly excluding zero) against the global PCC quantile and by -14.94M kWh (CI [-27.06M, -6.23M]) against Gaussian parametric reserve sizing. In transitional regimes (10% to 90% of turbines pitching), the joint posterior saves -1.48M kWh (CI [-2.14M, -1.07M]) over the continuous physical pitch rule, demonstrating that boundary-conditioned risk pricing retains substantial economic value after fleet-wide error cancellation.

VI. DISCUSSION

A. Accuracy, accountability and deployment gates

The accuracy cost is not secondary: iTransformer, Graph WaveNet, and lag baselines remain superior whole-sample forecasters on WTB; the routed model should pair with an established low-RMSE forecaster when aggregate accuracy is paramount. Its targeted role is an accountable operating-state diagnostic for boundary-specific decisions. External testing is graded rather than binary: La Haute Borne demonstrates anchor-observable replication (canonical NMI 0.975; withheld probe 0.674/0.575). On Kelmarsh (55% pitch) the probe reaches 0.340/0.378, and on Penmanshiel (78%) 0.302 (signature_core 0.195, far above permuted controls; Supplementary Table A9d). Partial pitch observability modulates rather than eliminates boundary recoverability. To evaluate operational durability against decadal aerodynamic wear and multi-year climate cycles, we conducted IEC 61400-12-1 density-calibrated walk-forward rolling evaluations across two independent commercial European wind farms spanning 17.6 cumulative turbine-operating years: Kelmarsh (9 full years, 2016–2024, 6 MM92 turbines, 473,184 steps) and Penmanshiel (8.6 full years, 2016–2024, 15 MM82 turbines, 451,334 steps). While static quantiles frozen at commissioning experience concept drift over multi-year horizons (crossing zero under decadal static evaluation), adopting standard utility two-year walk-forward rolling recalibration eliminates concept drift: on Kelmarsh, 6 out of 7 rolling folds (Folds 2–7, 2018–2024) strictly exclude zero with negative 95% bootstrap CIs, saving an average of -1.025 Million kWh/year ($p < 0.05$); on Penmanshiel, following initial commissioning, all 4 consecutive mature operational folds (Folds 3–6, 2020–2023) strictly exclude zero, saving an average of -2.029 Million kWh/year

TABLE VI
DEPLOYMENT CHECKS IDENTIFIED BY MULTI-FARM TESTING. THE
WITHHELD-CHANNEL PROBE REPLACES THE BINARY OBSERVABILITY GATE WITH A
GRADED SIGNATURE-STRENGTH READING.

Gate	Observed evidence	Required decision before reserve use
Withheld-channel signature	LHB 0.675; Penmanshiel 0.302 (core 0.195 borderline); Kelmarsh 0.340-0.378; shuffled controls at chance at every farm	Grade farms by signature strength; no binary observability no-go
Held-out routing criterion	Cross-farm held-out NMI 0.557 passes the 0.50 criterion in aggregate but is direction-dependent (0.341 in one direction)	Re-establish routing agreement locally and report per-direction values
Local boundary recalibration	Validation-only recalibration moves test NMI only marginally	Treat threshold transfer as insufficient
Pitch/proxy observability	Farms with partial pitch coverage recover the boundary above chance	Prefer direct pitch or a validated proxy for reserve-grade use
Geometry and regime support	Farm scale, sensor fields, power curves, and regime shares differ across sites	Run a local evidence protocol before gate-bin reserve allocation

($p < 0.05$). Together, 10 out of 11 mature annual folds strictly exclude zero across both commercial farms, proving that the jointly-learned boundary posterior delivers durable economic value under realistic utility maintenance protocols. Direct reserve deployment at new farms still requires pitch or proxy observability, boundary support, compatible geometry, local recalibration, and a held-out routing pass [18].

VII. LIMITATIONS

WTB pseudo-labels derive from wind speed and pitch angle, channels present in full gate anchors. Withheld-channel probes prove the boundary survives without them via consequence signatures across four farms (permuted controls at chance). Removing pitch dispersion retains non-power NMI 0.309 on WTB and 0.578 on La Haute Borne, confirming dispersion does not carry the signal. Yet routing remains an anchor-constrained diagnostic rather than anchor-free discovery [15], [17]. Inputs may contain history $\text{Patv}_{t-H+1:t}$ and issue-time Patv_t , while targets begin at $t + 1$. Expert semantics reflect the declared mapping; unassigned logits are not universal physical states. Comparisons are bounded to empirical-quantile screening, excluding optimal power flow, unit commitment, and market settlement [24], [26]. Strongest claims remain five-seed WTB recovery, mechanism intervention, La Haute Borne replication, and same-router reserve triage.

VIII. CONCLUSION

This study evaluated a jointly-learned boundary-risk posterior for wind-turbine reserve screening near the MPPT-to-pitch boundary, and localized its value with counterfactual controls. Evidence is consistent with shared representation via joint learning driving reserve risk pricing: the joint posterior beats the global quantile by 0.57M and an independent GBDT posterior by 1.6M, while a continuous physical pitch quantile remains the honest clean-observation baseline (15.538M). The routing structure provides degradation resilience: the routed posterior keeps 0.933 early-window recall at a six-step confirming-stream delay against 0.196 for the rule and 0.286 for a non-routed posterior head. The boundary stays identifiable when the label-defining channels are withheld (NMI

0.561, chance under permuted labels), and this consequence signature persists across external farms under heterogeneous observability (La Haute Borne, Penmanshiel, and Kelmarsh with 99%, 78%, and 55% pitch coverage) with negative controls everywhere, reflecting graded recoverability rather than a simplistic coverage law. Across 17.6 cumulative operating years on Kelmarsh and Penmanshiel, walk-forward rolling recalibration eliminates multi-year concept drift, yielding 10 mature annual evaluation folds strictly excluding zero (saving -1.03M and -2.03M kWh/yr). The clean-anchor rule and a simple classifier remain stronger clean detectors. The resulting system is an auditable defense-in-depth screening layer whose use depends on a graded signature-strength check and a held-out local pass, providing robust reserve triage when telemetry degrades without attempting to displace low-RMSE point forecasting or market dispatch.

IX. AI USE STATEMENT

The authors used OpenAI ChatGPT/Codex only for language editing, consistency checks, and submission-material drafting; all data, analyses, references, conclusions, and final text were reviewed and controlled by the authors.

X. CODE AND DATA AVAILABILITY

The raw KDD Cup 2022, ENGIE La Haute Borne, Kelmarsh, and Penmanshiel SCADA datasets are public; raw third-party data are not redistributed. Code, configurations, releasable derived tables, figure data, and checkpoints will be made available with the article. The reproduction package covers WTB routing, reserve audit, anchor-stress cache/train/guard, early-warning label degradation, classifier control, class-weight provenance/sensitivity, external diagnostics, and evidence-freeze protocols. Results are statistically reproducible across declared seeds rather than bitwise deterministic across all GPU/CUDA environments. The analysis involves no human subjects.

REFERENCES

- [1] P. Pinson, "Wind energy: Forecasting challenges for its operational management," *Statistical Science*, vol. 28, no. 4, pp. 564–585, 2013, doi: 10.1214/13-STS445.
- [2] R. Doherty and M. O'Malley, "A new approach to quantify reserve demand in systems with significant installed wind capacity," *IEEE Transactions on Power Systems*, vol. 20, no. 2, pp. 587–595, 2005, doi: 10.1109/TPWRS.2005.846206.
- [3] Y. Wang, F. Zhang, H. Kou, R. Zou, Q. Hu, and J. Wang, "A review of predictive uncertainty modeling techniques and evaluation metrics in probabilistic wind speed and wind power forecasting," *Applied Energy*, vol. 396, p. 126234, 2025, doi: 10.1016/j.apenergy.2025.126234.
- [4] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, "Graph WaveNet for deep spatial-temporal graph modeling," in *Proceedings of the 28th international joint conference on artificial intelligence*, 2019, pp. 1907–1913. doi: 10.24963/ijcai.2019/264.
- [5] J. Park and J. Park, "Physics-induced graph neural network: An application to wind-farm power estimation," *Energy*, vol. 187, p. 115883, 2019, doi: 10.1016/j.energy.2019.115883.
- [6] D. Kim, G. Ryu, and C. Moon, "Accuracy of a short-term wind power forecasting model based on deep learning using LiDAR-SCADA integration: A case study of the 400-MW Anholt offshore wind farm," *Applied Energy*, vol. 373, p. 123882, 2024, doi: 10.1016/j.apenergy.2024.123882.
- [7] S. Daenens, T. Verstraeten, P.-J. Daems, A. Nowé, and J. Helsen, "Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data," *Wind Energy Science*, vol. 10, pp. 1137–1152, 2025, doi: 10.5194/wes-10-1137-2025.
- [8] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," *Neural Computation*, vol. 3, no. 1, pp. 79–87, 1991, doi: 10.1162/neco.1991.3.1.79.

- [9] M. I. Jordan and R. A. Jacobs, "Hierarchical mixtures of experts and the EM algorithm," *Neural Computation*, vol. 6, no. 2, pp. 181–214, 1994, doi: 10.1162/neco.1994.6.2.181.
- [10] N. Shazeer *et al.*, "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in *International conference on learning representations*, 2017. Available: <https://openreview.net/forum?id=B1ckMDqIlg>
- [11] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *Journal of Machine Learning Research*, vol. 23, no. 120, pp. 1–39, 2022, Available: <https://www.jmlr.org/papers/v23/21-0998.html>
- [12] H. Hersbach *et al.*, "The ERA5 global reanalysis," *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999–2049, 2020, doi: 10.1002/qj.3803.
- [13] S. Guo, Y. Lin, N. Feng, C. Song, and H. Wan, "Attention based spatial-temporal graph convolutional networks for traffic flow forecasting," in *Proceedings of the AAAI conference on artificial intelligence*, 2019, pp. 922–929, doi: 10.1609/aaai.v33i01.3301922.
- [14] L. Bai, L. Yao, C. Li, X. Wang, and C. Wang, "Adaptive graph convolutional recurrent network for traffic forecasting," in *Advances in neural information processing systems*, 2020, pp. 17804–17815.
- [15] N. Zehtabiyani-Rezaie, A. Iosifidis, and M. Abkar, "Physics-guided machine learning for wind-farm power prediction: Toward interpretability and generalizability," *PRX Energy*, vol. 2, no. 1, p. 013009, 2023, doi: 10.1103/PRXEnergy.2.013009.
- [16] A. Karpatne *et al.*, "Theory-guided data science: A new paradigm for scientific discovery from data," *IEEE Transactions on Knowledge and Data Engineering*, vol. 29, no. 10, pp. 2318–2331, 2017, doi: 10.1109/TKDE.2017.2720168.
- [17] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, "Physics-informed machine learning," *Nature Reviews Physics*, vol. 3, no. 6, pp. 422–440, 2021, doi: 10.1038/s42254-021-00314-5.
- [18] J. Tautz-Weinert and S. J. Watson, "Using SCADA data for wind turbine condition monitoring: A review," *IET Renewable Power Generation*, vol. 11, no. 4, pp. 382–394, 2017, doi: 10.1049/iet-rpg.2016.0248.
- [19] E. A. Bossanyi, "The design of closed loop controllers for wind turbines," *Wind Energy*, vol. 3, no. 3, pp. 149–163, 2000, doi: 10.1002/we.34.
- [20] F. D. Bianchi, H. De Battista, and R. J. Mantz, *Wind turbine control systems: Principles, modelling and gain scheduling design*. London: Springer, 2006, doi: 10.1007/1-84628-493-7.
- [21] L. Y. Pao and K. E. Johnson, "Control of wind turbines: Approaches, challenges, and recent developments," *IEEE Control Systems Magazine*, vol. 31, no. 2, pp. 44–62, 2011, doi: 10.1109/MCS.2010.939962.
- [22] K. E. Johnson, L. J. Fingersh, M. J. Balas, and L. Y. Pao, "Methods for increasing Region 2 power capture on a variable-speed wind turbine," *Journal of Solar Energy Engineering*, vol. 126, no. 4, pp. 1092–1100, 2004, doi: 10.1115/1.1792653.
- [23] E. Ela, M. Milligan, and B. Kirby, "Operating reserves and variable generation," National Renewable Energy Laboratory, Golden, CO, NREL/TP-5500-51978, 2011, doi: 10.2172/1023095.
- [24] J. B. Bremnes, "Probabilistic wind power forecasts using local quantile regression," *Wind Energy*, vol. 7, no. 1, pp. 47–54, 2004, doi: 10.1002/we.107.
- [25] Y. Zhang, J. Wang, and X. Wang, "Review on probabilistic forecasting of wind power generation," *Renewable and Sustainable Energy Reviews*, vol. 32, pp. 255–270, 2014, doi: 10.1016/j.rser.2014.01.033.
- [26] Z. Zhou *et al.*, "Application of probabilistic wind power forecasting in electricity markets," *Wind Energy*, vol. 16, no. 3, pp. 321–338, 2013, doi: 10.1002/we.1496.
- [27] Y. Li, R. Yu, C. Shahabi, and Y. Liu, "Diffusion convolutional recurrent neural network: Data-driven traffic forecasting," in *International conference on learning representations*, 2018. Available: <https://openreview.net/forum?id=SJiHXGWAZ>
- [28] H. A. Nielsen, H. Madsen, and T. S. Nielsen, "Using quantile regression to extend an existing wind power forecasting system with probabilistic forecasts," *Wind Energy*, vol. 9, no. 1–2, pp. 95–108, 2006, doi: 10.1002/we.180.
- [29] J. Zhou *et al.*, "SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array," *Scientific Data*, vol. 11, p. 649, 2024, doi: 10.1038/s41597-024-03427-5.
- [30] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," in *International conference on learning representations*, 2023. Available: <https://openreview.net/forum?id=Jbdc0vTOcol>
- [31] Y. Liu *et al.*, "iTransformer: Inverted transformers are effective for time series forecasting," in *International conference on learning representations*, 2024. Available: <https://openreview.net/forum?id=JePfAI8fah>
- [32] A. Das, W. Kong, A. Leach, and R. Sen, "Long-term forecasting with TiDE: Time-series dense encoder," in *Transactions on machine learning research*, 2023. Available: <https://openreview.net/forum?id=pCbC3aQB5W>